

Alireza Ghorbanzadeh

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EDUCATION

McMaster University

Sep. 2018 – Apr. 2028

Bachelor of Engineering (Biomedical Engineering and Health Sciences)

Hamilton, Ontario

- **Related Coursework:** IBEHS 4P03(Health Solution Design Project IV), COMPENG 2DX3(Microprocessors), ELECENG 3EJ4(circuits and devices II), ELECENG 4BF4(advanced medical imaging)

EXPERIENCE

McMaster University

Sep. 2026 – Present

Teaching Assistant – IBEHS 3S03

Hamilton, Ontario

- Lead **hands-on Python and JupyterLab tutorials** focused on statistical analysis of biomedical data, guiding students through implementation, interpretation, debugging, and quantitative problem-solving.
- Develop and refine computational tools that translate statistical concepts into **practical biomedical applications** and reproducible analysis workflows.

AliThriftyCo.

Dec. 2020 – Nov. 2022

Founder and Marketing Director

Toronto, Ontario

- Designed, built, and iteratively tested **biomedical and electrical prototypes**, including an **EMG-based extraocular muscle-signal acquisition system** integrating **custom surface electrodes, analog circuitry, and microcontrollers**.
- Redesigned and adapted a kitchen roll-holder product for the North American market using **CAD**, coordinated production with overseas suppliers, and successfully **sold 300+ units**
- Developed marketing and educational content that **raised awareness of ocular conditions**, including strabismus.

McMaster University

Sep. 2020 – Dec. 2020

Teaching Assistant – IBEHS 1P13

Hamilton, Ontario

- Mentored first-year engineering students in **CAD design and Python programming**, providing hands-on technical guidance throughout multidisciplinary design projects.
- Guided student teams through open-ended health technology design projects and supported **iterative solution development**.

Extracurricular Involvement and Projects

McMaster Design League - Design Competitor

Jan 2026

- Collaborated on the design and rapid prototyping of a toy race car, contributing to the development of a functional solution under competition constraints.
- Applied an **iterative engineering design process**, moving from concept generation and CAD/design decisions to fabrication, testing, and refinement of the final prototype.

DIY Dynamic Webcam

Jan 2021

- Designed and built a dynamic webcam system using an ESP32-CAM integrated into a custom **3D-printed enclosure** inspired by the **human eye**
- Engineered a **two-axis pan-and-tilt mechanism** using stepper motors, enabling independent horizontal and vertical camera movement.

SKILLS

Programming : Python, Embedded C, JupyterLab, HTML, CSS, JavaScript

Prototyping and Hardware : CAD, 3D printing, circuit design, Microcontrollers, Rapid prototyping, Hardware testing